Machine Intelligence and Learning Systems

# Deep learning algorithms for large and robust models


- **Keywords :** Deep learning, Adversarial Attack, Randomized Smoothing

- **Duration :** 4 à 6 mois.

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**Context :** Deep learning algorithms have shown their vulnerability to adversarial attacks [4], small and imperceptible perturbations of the inputs that maliciously fool neural networks. Since this discovery, building adversarial attacks and defenses against them has been an active and hot research topic. Randomized smoothing is a widely recognized defense mechanism against adversarial attacks, providing state-of-the-art robustness for output predictions, [2, 3, 1]. This method, known for its certificates, involves convolving a classifier with a Gaussian distribution. However, the development of this method remains in the early stages as there is still room for improving the randomized smoothing procedure.

**Goals :** This internship project aims to investigate the efficacy of randomized smoothing during the generation of certificates for smoothed classifiers. The process involves the application of additive Gaussian noise to the neural network at inference, resulting in the estimation of the smoothed classifier through Monte Carlo integration. For instance, emphasis could be placed on controlling the inherent variance introduced by Monte Carlo integration to ensure the production of a robust and conservative certificate. This is crucial as new top architectures such as Transformers have an expensive inference cost.

**Organisation :** The internship will start with a survey of the literature on adversarial attacks with a focus on recent papers dedicated to randomized smoothing. The experimental framework will use the python library PYTORCH.

## Références


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[2] Jeremy M. Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified Adversarial Robustness via Randomized Smoothing, June 2019. arXiv :1902.02918 [cs, stat].

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[4] Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. Intriguing properties of neural networks. *arXiv preprint arXiv :1312.6199*, 2013.